

Lecture 4 (Additional Notes)

Recall the Lemma from Lecture 2:

Lemma: Suppose $L/\mathbb{Q}$ is an abelian extension of degree n . For any prime q such that $q|\delta_L$, but $q \nmid n$, there is an abelian extension $F/\mathbb{Q}$ such that the degree of $[F : \mathbb{Q}]$ divides n , $L \subseteq F(\zeta_q)$, $q \nmid \delta_F$, and if q' is a prime dividing δ_F , then q' divides δ_L as well.

A sketch of the argument is below.

First, assume that L contains a primitive q th root of unity so that $\mathbb{Q} \subseteq \mathbb{Q}(\zeta_q) \subseteq L$ (and let $K = \mathbb{Q}(\zeta_q)$). Let $\tilde{Q}$ be a prime of $\mathcal{O}_L$ such that $\tilde{Q} \cap \mathbb{Z} = q\mathbb{Z}$. One first shows that $L = L^{V_1}$, where V_1 is the first ramification group for $\tilde{Q}$ in $\text{Gal}(L/\mathbb{Q})$. Then one shows (by comparing ramification indices to corresponding degrees of subextensions) that $L/\mathbb{Q}(\zeta_q)$ is unramified for the prime ideal ideal of $\mathcal{O}_K$, $\tilde{Q} \cap \mathcal{O}_K$. You then show that L^I , the inertial field of $\tilde{Q}$ for $L/\mathbb{Q}$ satisfies the conditions by showing that $L = L^I(\zeta_q)$ has degree dividing n and has $q \nmid \delta_{L^I}$. (The containment argument come down to showing determining the inertial field of $\tilde{Q}$ for L/K as well as showing that the field fixed by the first higher ramification group of $\tilde{Q}$ for L/K is the compositum of K and L^{V_1} . A degree calculation then lets you show that $L \subseteq L^I(\zeta_q)$. The discriminant argument comes down to the fact that q is unramified in L^I , and if q' divides δ_{L^I} , it also divides δ_L .)

For the general case, where L does not contain ζ_q , one uses the compositum $K = L(\zeta_q)$ and an inertial field $F = K^I$ satisfying $L \subseteq F = K(\zeta_q)$.